

2016.0 RANGE ROVER (LG), 412-01

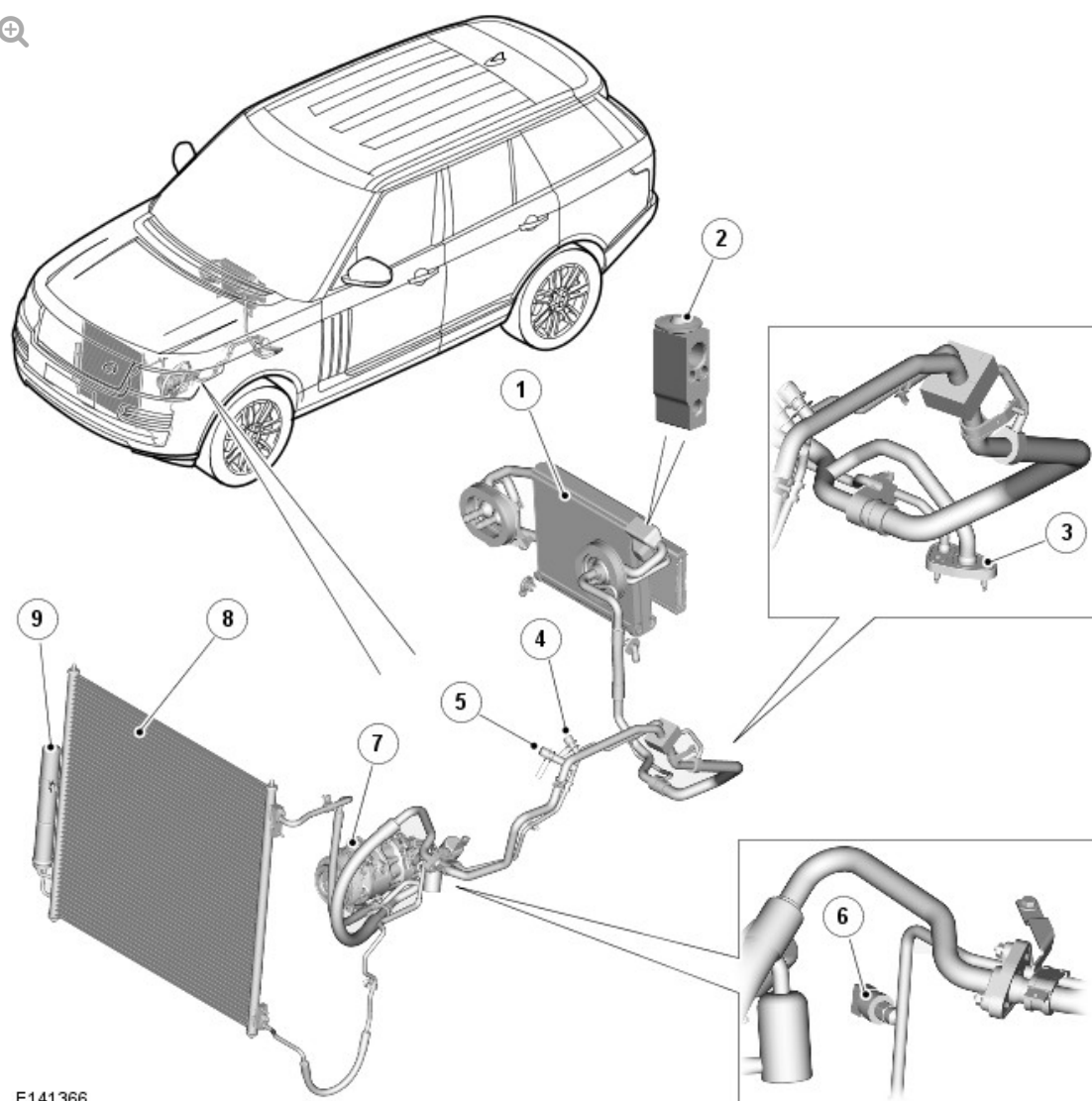
CLIMATE CONTROL

DESCRIPTION AND OPERATION

COMPONENT LOCATION

NOTE:

Installation on 5.0L V8 petrol engine shown, other installations similar.



E141366

ITEM	DESCRIPTION
1	Evaporator
2	Thermostatic expansion valve
3	Air conditioning connections for auxiliary climate control (where fitted)
4	Low pressure servicing connection
5	High pressure servicing connection
6	Refrigerant pressure sensor
7	Air conditioning compressor
8	Condenser
9	Receiver drier

OVERVIEW

The A/C (air conditioning) system is a sealed, closed loop system filled with a charge weight of refrigerant as the heat transfer medium. Depending on market, the refrigerant is either R1234yf or R134a. Oil is added to the refrigerant to lubricate the internal components of the A/C compressor. The system consists of:

- An A/C compressor
- A condenser
- A receiver drier
- A thermostatic expansion valve
- An evaporator
- Low and high pressure refrigerant lines.

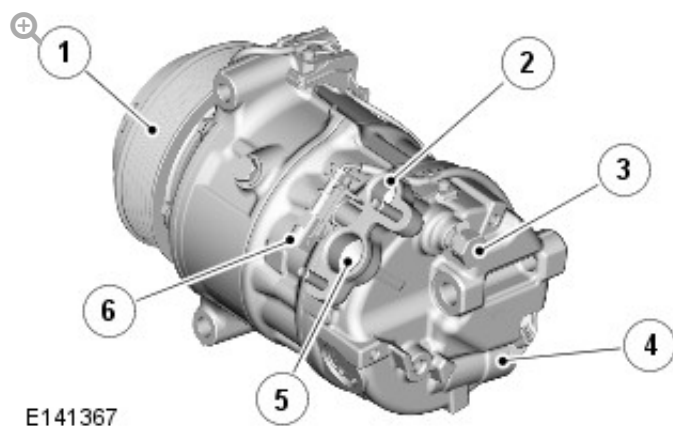
A pressure sensor in the high pressure refrigerant line allows the ATC (automatic temperature control) module to monitor the status of the A/C system.

For additional information, refer to: [Control Components](#) (412-01A Climate Control, Description and Operation).

DESCRIPTION

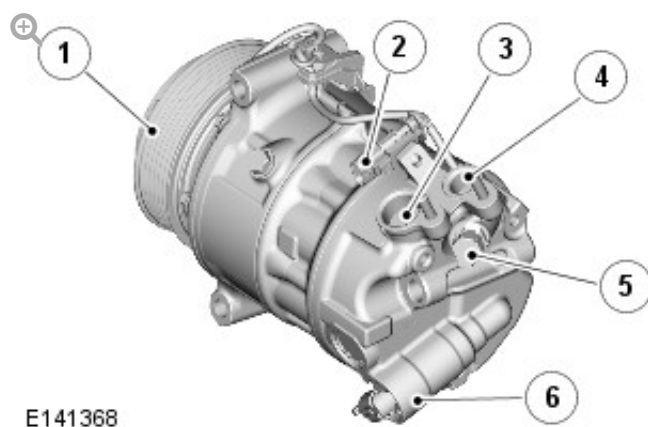
AIR CONDITIONING COMPRESSOR

V8 5.0L Petrol, V8 S/C 5.0L Petrol and TDV8 4.4L Diesel



ITEM	DESCRIPTION
1	Pulley
2	Refrigerant outlet port
3	Pressure relief valve
4	Electronic control valve
5	Refrigerant inlet port
6	Electrical connector

TDV6 3.0L Diesel



ITEM	DESCRIPTION
1	Pulley
2	Electrical connector

3	Refrigerant inlet port
4	Refrigerant outlet port
5	Pressure relief valve
6	Electronic control valve

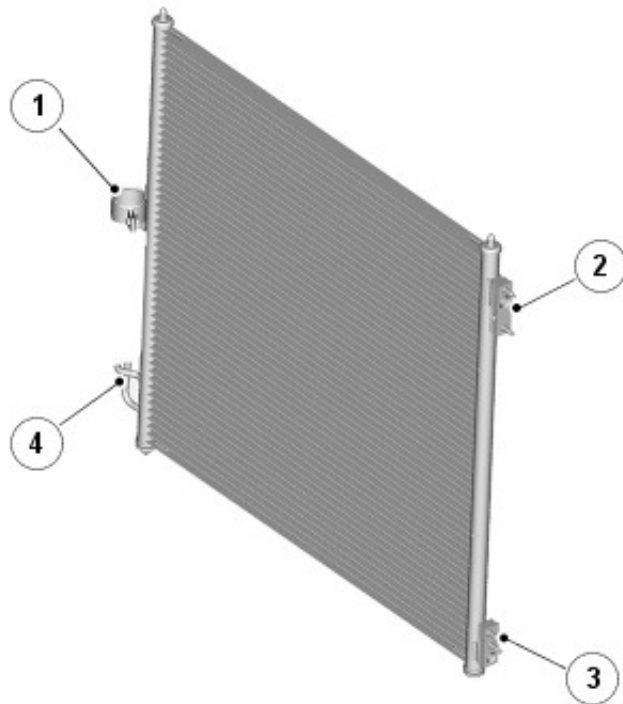
On all engine variants, the A/C compressor is a variable displacement unit attached to the front left side of the engine with three bolts.

The compressor is driven by the engine accessory drive belt via an electro-magnetic clutch in the compressor pulley. The ATC module controls the operation of the electro-magnetic clutch. The clutch incorporates a thermal cut-off fuse, which disconnects the power feed from the ATC module if the temperature increases to $182 \pm 5 \text{ }^{\circ}\text{C}$ ($360 \pm 9 \text{ }^{\circ}\text{F}$).

The displacement of the compressor is controlled by an integral electronic control valve, which is operated by the ATC module. The ATC module automatically adjusts the displacement of the A/C compressor, between the minimum and maximum values, to match the thermal load of the evaporator.

To protect the refrigerant system from excessive pressure, a pressure relief valve is installed in the outlet side of the A/C compressor. The pressure relief valve vents excess pressure into the engine compartment.

CONDENSER



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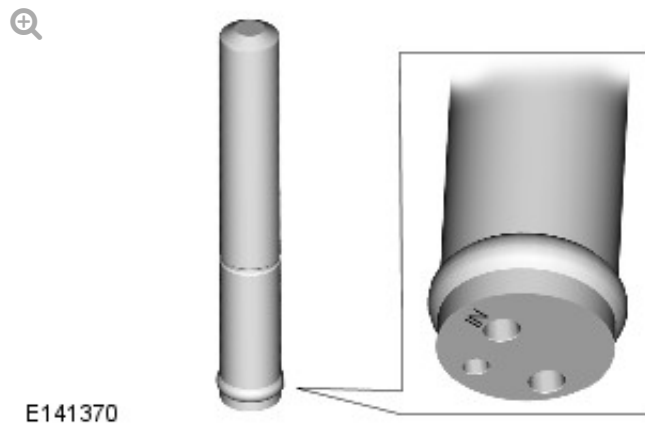
ITEM	DESCRIPTION
1	Receiver drier attachment bracket
2	Refrigerant inlet port
3	Refrigerant outlet port
4	Receiver drier pipes

The condenser is installed immediately in front of the radiator. Pins on the bottom of the end tanks locate in bushes installed in brackets integrated into the bottom of the radiator. Two brackets attach pins on the top of the end tank to pins on the top of the radiator.

The condenser is classified as a sub-cooling condenser and consists of a fin and tube heat exchanger core installed between two end tanks. Divisions in the end tanks separate the heat exchanger into a four pass upper (condenser) section and a two pass lower (sub-cooler) section. Two

connector blocks on the left end tank of the condenser provide connections for the high pressure lines from the A/C compressor and the evaporator. Two pipes at the bottom of the right end tank of the condenser provide connections for the receiver drier.

RECEIVER DRIER



The receiver drier is attached to the two stub pipes on the right end tank of the condenser. A collar, located on lands on the stub pipes and secured with a bolt, attaches the stub pipes to the receiver drier. The body of the receiver drier is installed in a bracket welded to the right end tank of the condenser.

The inlet and outlet ports of the receiver drier are the same size, so care must be taken to install the receiver drier the correct way round on the stub pipes. To assist with installation, the inlet port is identified with the word IN etched into the receiver drier.

Refrigerant entering the receiver drier passes through a filter and a desiccant pack, then collects in the base of the unit before flowing through the outlet stub pipe back to the condenser. The desiccant and the filter are non-serviceable; the complete unit must be replaced when a change of desiccant is required.

THERMOSTATIC EXPANSION VALVE

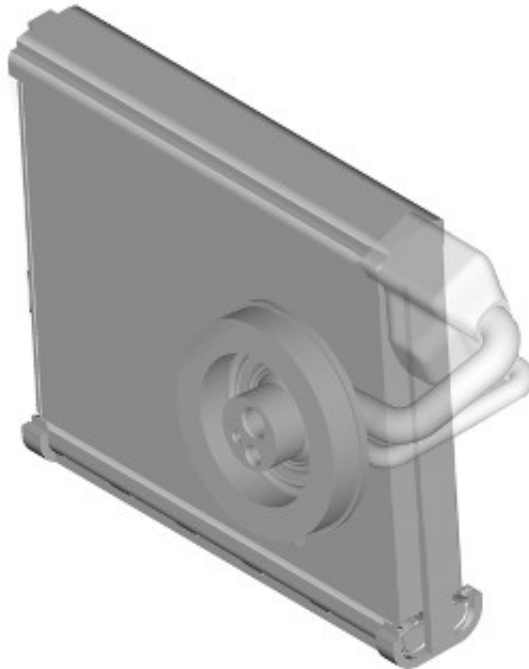


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The thermostatic expansion valve is a block type valve attached to the inlet and outlet ports of the evaporator, under a thermal insulation cover.

Two bolts secure the refrigerant pipes and the thermostatic expansion valve to the inlet and outlet ports of the evaporator.

EVAPORATOR



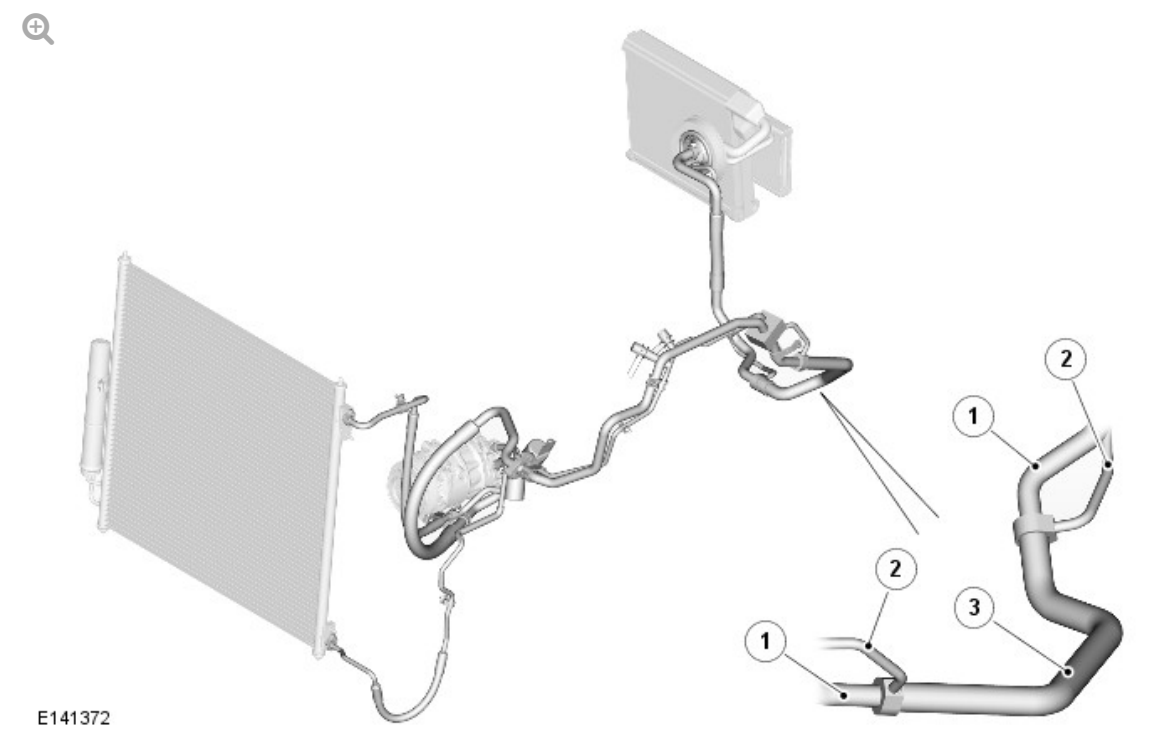
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The evaporator is a fin and tube heat exchanger installed in the heater assembly, between the blower and the heater core.

REFRIGERANT LINES

NOTE:

Installation on 5.0L V8 petrol engine shown, other installations similar.



ITEM	DESCRIPTION
1	Low pressure pipe
2	High pressure pipe
3	Internal heat exchanger

The refrigerant lines consist of a combination of rigid pipes and flexible hoses that connect the thermostatic expansion valve on the evaporator to the A/C compressor and the condenser. An internal heat exchanger increases the efficiency of the evaporator and ensures any residual liquid in the low pressure line is evaporated before it reaches the compressor.

NOTE:

The internal heat exchanger is incorporated primarily because of the introduction of refrigerant R1234yf (which is less efficient than refrigerant R134a), but is included in all systems for commonality reasons.

Low and high pressure servicing connections are incorporated into the refrigerant lines for system servicing. The high pressure line also incorporates a refrigerant pressure sensor for the climate control system. For additional information, refer to: [Control Components](#) (412-01A Climate Control, Description and Operation).

On vehicles with auxiliary climate control, connections for the auxiliary A/C system are incorporated into the refrigerant lines in the left rear corner of the engine compartment.

For additional information, refer to: [Auxiliary Climate Control](#) (412-02A Auxiliary Climate Control, Description and Operation).

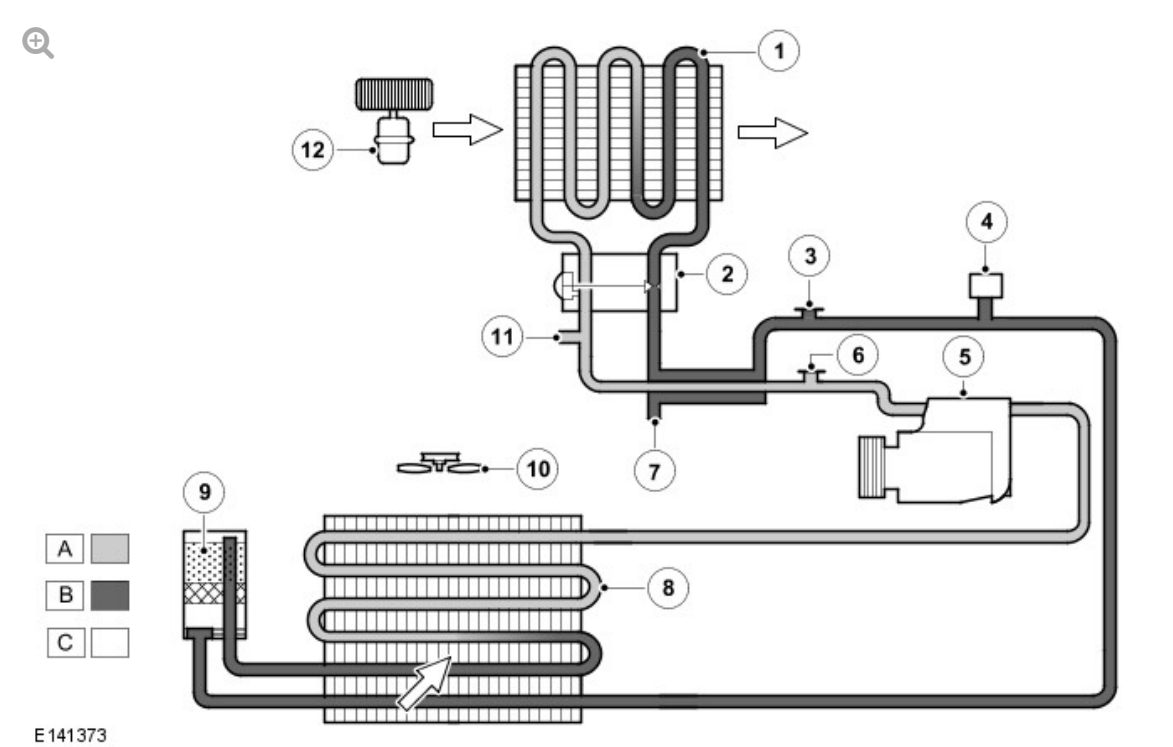
OPERATION

The A/C compressor circulates the refrigerant around the system by compressing low pressure, low temperature vapor from the evaporator and discharging the resultant high pressure, high temperature vapor to the condenser. In the condenser the vapor converts to a liquid, which then passes through the receiver drier and out of the condenser to the thermostatic expansion valve. As it passes through the thermostatic expansion valve, the liquid refrigerant changes to a fine, low pressure, spray, which is directed into the evaporator. In the evaporator the refrigerant changes back to a low pressure, low temperature vapor, as it absorbs heat from the air in the passenger compartment, then returns to the A/C compressor to begin the cycle again.

Operation of the air conditioning system is controlled by the ATC module,

which adjust the electronic control valve in the A/C compressor to match the refrigerant flow around the system to the thermal load of the evaporator. By matching refrigerant flow to the thermal load of the evaporator, the ATC module maintains the required temperature in the passenger compartment while maximizing fuel economy.

Air Conditioning System Schematic



A = REFRIGERANT VAPOR; B = REFRIGERANT LIQUID; C = AIR FLOW.

ITEM	DESCRIPTION
1	Evaporator
2	Thermostatic expansion valve
3	High pressure servicing connection
4	Refrigerant pressure sensor
5	Air conditioning compressor
6	Low pressure servicing connection
7	High pressure connection for auxiliary climate control (where fitted)
8	Condenser

9	Receiver drier
10	Engine cooling fan(s)
11	Low pressure connection for auxiliary climate control (where fitted)
12	Blower